A Study on Speech Emotion Recognition Based on Multi-Feature and Deep Models

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Abstract

This dissertation addresses the critical challenge of robust Speech Emotion Recognition (SER) in the presence of environmental noise. While deep learning has significantly advanced SER, model performance often degrades when moving from clean, laboratory-recorded data to more realistic, noisy conditions. This study conducts a systematic evaluation of deep learning models and audio features to improve noise robustness and generalization. We investigate four distinct architectures: a Multilayer Perceptron (MLP), a shallow Convolutional Neural Network (CNN), a deeper ResNet-18, and the attention-based Audio Spectrogram Transformer (AST). These models are trained on both traditional Mel-frequency cepstral coefficient (MFCC) features and 2D Mel-spectrograms.

The core of our methodology revolves around a comprehensive on-the-fly data augmentation strategy, where training data is mixed with white noise and natural soundscapes from the ESC-50 dataset at various signal-to-noise ratios (SNRs). We evaluate model performance on two standard benchmarks, RAVDESS and IEMOCAP, under both clean and noisy test conditions. The findings consistently demonstrate that the AST architecture surpasses the convolutional and MLP-based models, particularly when trained with data augmentation. Augmentation not only preserves but often improves AST's performance on clean test sets while significantly enhancing its robustness against a wide range of noises and SNRs.

Furthermore, our analysis reveals a crucial interaction between model architecture and augmentation strategy. While AST effectively leverages diverse acoustic perturbations to learn noise-invariant features, the ResNet-18 model proves more fragile, exhibiting performance degradation under the same conditions. The study also highlights dataset-specific effects, where augmentation leads to a more balanced precision-recall trade-off on the heterogeneous IEMOCAP dataset, particularly for the AST model. Ultimately, this research underscores the synergy between transformer architectures and robust data augmentation, presenting a promising framework for developing more practical and generalizable SER systems.

Acknowledgements

Time has flown by, and my master's journey has now reached its conclusion. Along this path, I have been fortunate to meet many people and experiences that I will always cherish.

First and foremost, I would like to express my deepest gratitude to my supervisor, Dr. Rishiraj Bhattacharyya, for his patient guidance and invaluable support throughout my research and writing. His calm attitude and rigorous academic spirit have not only provided me with direction but also given me the confidence to overcome challenges along the way.

I am equally grateful to all the sincere and wonderful friends I met during my year in the UK. You made life in a foreign country warm and memorable, turning ordinary days into something bright with kindness and laughter.

I would also like to thank myself—for making the choice to pursue this path and for holding on with persistence. And to my parents, whose understanding and unconditional support have always been my strongest foundation, I owe more than words can express.

Life is a continuous journey forward, and the future shines with light. May we all carry hope in our hearts, live fully in the present, and embrace the bright days ahead.

Declarations

I certify that this project is my own work. Code development was assisted by Claude Sonnet 3.7 and debugging by ChatGPT 4.0. The report structure and initial drafts are mine; final editing was assisted by ChatGPT for clarity and consistency. All outputs have been reviewed and edited to accurately reflect my work.

Abbreviations

AST	Audio Spectrogram Transformer
CCC	concordance correlation coefficient
CNN	convolutional neural network
DCT	discrete cosine transform
DFT	discrete Fourier transform
DTs	Decision Trees
FFN	Feed-forward network
GMMs	Gaussian Mixture Models
HMMs	Hidden Markov Models
HSFs	high-level functional statistics
KNN	K-nearest neighbors
LLDs	low-level descriptors
MFCCs	Mel-frequency cepstral coefficients
MHSA	multi-head self-attention
MLP	multilayer perceptron
RMS	root-mean-square
SER	Speech Emotion Recognition
SNR	signal-to-noise ratio
SVMs	Support Vector Machines
TEO	Teager Energy Operator
UAR	unweighted average recall
ViT	Vision Transformer

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CHAPTER 1

Introduction

1.1 Background

Speech conveys affective information via prosody, articulation, and spectral cues, enabling inference of speakers' intentions and internal states. Speech Emotion Recognition (SER) infers emotion from acoustic signals as discrete categories (e.g., anger, happiness, sadness, fear, neutral) or continuous dimensions (e.g., valence, arousal, dominance) [13]. Rooted in speech processing, affective computing, and psychology, early SER systems relied on handcrafted prosodic and spectral features with classical classifiers [17]. Over the past decade, deep learning—CNNs on time–frequency representations, recurrent and attention-based models for temporal dynamics, and transformer architectures with self-supervised pretraining—has markedly improved generalization [17, 46, 43, 15, 3]. The focus has shifted from acted corpora to in-the-wild data, emphasizing robustness to noise, channel variability, and cultural–linguistic diversity. Attention to transparency, privacy, and fairness has also increased [16].

SER has broad practical value. In customer service and contact centers, emotion-aware analytics can monitor satisfaction and support agent coaching in real time [39]. In healthcare and mental health, non-invasive acoustic markers assist in screening and monitoring of affective disorders and stress. In human–computer interaction, assistants and social robots adapt responses and dialogue strategies to users' emotional cues [14]. In automotive and smart environments, continuous monitoring of driver frustration or fatigue enhances safety and personalization. These applications motivate solutions that are accurate, low-latency, and resource-efficient for cloud and on-device deployment.

The research significance of SER is twofold. Scientifically, SER advances understanding of how affect is encoded in speech and how to learn robust, interpretable representations under limited, imbalanced, and noisy supervision [22, 23]. Technically, it drives innovation in domain generalization, cross-lingual transfer, and multimodal fusion, while promoting explainability [1]. Addressing ethical and legal considerations—including data privacy, informed consent, and bias mitigation—is essential to responsible deployment. Finally, rigorous benchmarking, reproducibility, and standardized evaluation protocols (e.g., accuracy metrics for categorical tasks and concordance correlation for continuous estimation) are critical for cumulative progress. Altogether, SER research provides theoretical insights and practical tools for empathetic, human-centered intelligent systems.

1.2 Main Contents of the Dissertation

This dissertation addresses suboptimal recognition accuracy in SER by developing a deep learning framework that integrates preprocessing, feature engineering, attention mechanisms, and model optimization. We systematically study SER methods, including audio preprocessing, data augmentation, feature selection, model architectures, and evaluation metrics. Our objective is to learn feature representations with stronger affective discriminability, thereby improving accuracy, robustness, and generalization across speakers, recording conditions, and datasets. The proposed approach emphasizes practical deployability by balancing performance with computational efficiency for cloud and on-device inference.

1.3 Structure of the Dissertation

The dissertation is organized as follows. Chapter 1 introduces the research background and motivation, identifies the key challenges in SER, and outlines the scope and structure of the thesis. Chapter 2 surveys related work on speech signal processing, feature extraction, the design of attention mechanisms, and emotion taxonomies and recognition paradigms, summarizing major international developments. Chapter 3 presents the methodology. We construct features from audio data using Mel-frequency cepstral coefficients (MFCCs) and Mel-spectrograms; we then investigate multiple models for SER, including a multilayer perceptron (MLP), a shallow convolutional neural network (CNN), ResNet-18 for 2D Mel-spectrogram inputs, and the Audio Spectrogram Transformer (AST), a ViT-based model for audio classification. Finally, we apply data augmentation by injecting noise during training to improve generalization and mitigate overfitting. Chapter 4 describes the experiments. We design comparative studies to evaluate the performance differences among features and models on SER tasks under consistent protocols and metrics. Chapter 5 provides discussion. We analyze the experimental results in depth, examine the relative strengths and weaknesses of different features and models, and articulate limitations and directions for future research. Chapter 6 concludes the thesis by summarizing the main findings and contributions, and highlighting potential avenues for subsequent work.

CHAPTER 2

Literature Review

2.1 Datasets for Speech Emotion Recognition

Datasets are foundational for SER and are broadly categorized into acted and in-the-wild corpora. Acted datasets are recorded in controlled laboratories with professional actors. They offer clean signals and clear affective cues but may over-represent exaggerated expressions and miss everyday subtleties. By contrast, spontaneous corpora capture naturalistic affect in real interactions, better reflecting deployment scenarios but introducing background noise, speaker overlap, label ambiguity, and class imbalance. Below we summarize several widely used datasets.

RAVDESS. The Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) [31] includes 24 actors (12 female, 12 male) uttering two matched statements across eight emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprise) with normal/strong intensities (neutral has no intensity), totaling 1,440 studio-recorded files. Balanced classes and clear cues suit benchmarking, though the acted style can limit ecological validity on spontaneous speech.

IEMOCAP. IEMOCAP [8] is a 12-hour multimodal corpus with 10 actors across five sessions, combining scripted and improvised dialogues. Utterances have multi-rater labels for nine emotions and three dimensions (valence, arousal, dominance). Many studies merge excitement into happiness, yielding four- or six-class protocols; we adopt both for comparability.

EmoDB. The Berlin Database of Emotional Speech [7] records 10 German actors producing 10 sentences across seven emotions: anger, sadness, happiness, fear, neutral, disgust, and boredom. After screening, about 500 studio utterances remain, providing clear emotional prototypes.

MELD. The Multimodal EmotionLines Dataset [36], derived from "Friends," adds audio and video to text, with over 1,400 conversations and 13,000 utterances annotated with seven emotions. Its multimodal, conversational structure enables contextual emotion modeling and speaker interaction analysis.

2.2 SER with Acoustic Features

Acoustic feature engineering compresses waveforms into descriptors from time- and frequency-domain analyses. Canonical categories include prosodic (pitch, energy, duration), voice-quality (jitter, shimmer, harmonicity), spectral (formants, cepstra), and TEO-based features. Low-level descriptors (LLDs) capture frame-wise micro-variations and are summarized into high-level statistics (means, derivatives, moments) to encode utterance-level context.

Empirical studies have shown that acoustic features carry substantial affective information, reflecting changes in emotion, speaker identity, speaking style, and tempo [30]. Lee et al. [24] and Schuller et al. [44] reported strong results using feature sets combining energy-, pitch-, and spectral-related cues, with pitch-related features often proving more discriminative than energy features. Rao et al. [38] compared LLDs against HSFs (and their combination) and found that combining frame-level prosody (e.g., F0 trajectories, energy) with statistical summaries improves recognition performance. Lugger et al. integrated prosodic and voice-quality features (e.g., glottal gradient, skewness gradient, closure rate gradient) within a two-stage classifier to reach 74.5% accuracy. Li et al. [27] further augmented HSFs with jitter and shimmer to boost performance. Zhang et al. [55] combined prosodic with voice-quality features (including the first three formants), improving accuracy by about 10% over prosody-only baselines. Sato et al. [42] used clustering to label frames and constructed a segmented MFCC feature set, outperforming K-nearest neighbors (KNN) on prosody and Hidden Markov Models (HMMs) on conventional MFCCs. Bitouk et al. [6] introduced new spectral features by computing MFCCs separately for stressed vowels, unstressed vowels, and consonants, concluding that consonants may carry richer affective cues than vowels. Zhou et al. [56] proposed three TEO variants—especially a standard-band TEO autocorrelation envelope—to study airflow energy changes under stress, achieving better results than pitch+MFCC combinations.

Classifiers then map features to affect. Traditional approaches include Gaussian Mixture Models (GMMs), Support Vector Machines (SVMs), and Decision Trees (DTs). Recent methods employ deep learning to learn discriminative representations from acoustic features, such as 1D CNNs and RNNs.

Overall, acoustic-feature-based SER offers interpretability and relatively low complexity: extraction is algorithmic with few learnable parameters, while deep models on these features improve adaptability and reduce overfitting. Challenges remain: STFT's global nature may miss non-stationary, localized patterns; attention on handcrafted features may fail to emphasize truly salient regions.

2.3 SER with Spectrogram-based Features

A central question in spectrogram-based SER is how to extract discriminative representations from time–frequency images. Mel-spectrograms preserve localized structure and delegate hierarchical feature learning to deep networks. Representing energy over time and frequency in 2D enables learning of complex spectro-temporal patterns that fixed handcrafted descriptors struggle to encode [34]. Networks can discover filter-like patterns akin to Mel filterbanks while capturing long-range dependencies and contextual cues.

Deep models learn affect representations without manual feature selection. Convolutional neural networks (CNNs) are particularly impactful in SER, as spectrograms exhibit strong local correlations along time and frequency; local receptive fields enable progressive feature abstraction from 2D inputs [37]. Recurrent neural networks (RNNs) and Long Short-Term Memory (LSTM) networks are commonly stacked on CNN features to model temporal dependencies and can also operate directly on frame-level acoustic features [4]. Attention mechanisms [47] further refine representations by emphasizing salient regions.

Trigeorgis et al. [46] and Lim et al. [29] proposed CNN–LSTM hybrids where CNNs extract spectrogram features and LSTMs capture sequence dynamics, outperforming traditional methods; similar architectures are widely adopted [41, 46]. Capsule networks have also been explored to model spectral spatial relations, achieving promising sentence-level results [48]. Representative architectures are summarized in 2.1.

Table 2.1: Related research on SER using spectrogram-based models

Study	Year	Method	Dataset	Result
Zou et al. [9]	2014	DBN	ABC Corpus	Accuracy: 52.2%
Kim et al. [19]	2019	DNN	IEMOCAP	UAR: 61.4%
Aldeneh et al. [2]	2017	CNN	IEMOCAP; MSP-IMPROV	UAR: 52.6%
Lee et al. [25]	2015	BiLSTM	IEMOCAP	UAR: 63.89%
Latif et al. [21]	2019	CNN-LSTM	IEMOCAP; MSP-IMPROV	UAR: 60.23%; 52.43%
Bhosale et al. [5]	2020	ACNN	IEMOCAP	UAR: 63.15%
Mustaqeem et al. [40]	2020	CNN-LSTM	EMODB	Accuracy: 85.57%

On top of high-capacity encoders, attention mechanisms are extensively used—including self-attention [47], local attention [32], hop-wise attention [53], and other variants [33, 52]. These modules emphasize informative time–frequency regions while suppressing irrelevant ones; see Table 2.2 for recent studies.

Table 2.2: Related research on SER using attention-based mechanisms

Study	Year	Method	Dataset	Result
Mirsamadi et al. [32]	2017	Local Attention	IEMOCAP	UAR: 58.8%
Li et al. [28]	2019	Self Attention	IEMOCAP	UAR: 58.1%
Yoon et al. [53]	2019	Multi-Hop Attention	IEMOCAP	UAR: 65.2%
Xie et al. [49]	2019	TF Attention	CASIA; eNTERFACE	UAR: 63.89%; 89.6%
Tarantino et al. [45]	2019	Self Attention	IEMOCAP	UAR: 63.8%
Xu et al. [50]	2020	Alignment Attention	IEMOCAP	UAR: 57.4%
Li et al. [26]	2021	Self BiAttention	EMODB	Accuracy: 85.95%

Finally, learned representations are mapped to task outputs via suitable losses and metrics. For categorical SER, cross-entropy remains dominant; mean squared error is often used for dimensional regression. As class imbalance grew, the Interspeech 2009 Emotion Challenge popularized unweighted average recall (UAR), now widely adopted. For dimensional SER, the concordance correlation coefficient (CCC) is common. Despite gains, challenges remain: fixed-size kernels can limit global context; attention heads may underuse local/global/positional cues; cross-entropy is sensitive to label distributions; variable-length utterances complicate segment-level training and alignment.

CHAPTER 3

Methodology

3.1 Feature Extraction

Feature extraction plays a crucial role in speech emotion recognition. A variety of acoustic features can be derived from emotional speech to reflect characteristics of a speaker's affective behavior. High-quality and informative features have a significant impact on recognition accuracy. How to extract and select features that effectively capture emotional variation remains one of the most important problems in the current field of speech emotion recognition.

This section details the feature extraction pipeline and unified notation used throughout the thesis. Let $x[n]$ denote the discrete-time speech signal sampled at rate f_s (Hz). Frames are indexed by m, frequency bins by k, and Mel bands by f. The frame length is N samples, hop size is H samples, the analysis window is $w[n]$ (Hann unless otherwise stated), the discrete Fourier transform (DFT) length is K, and the number of Mel filters is F. We use $X[k, m]$ for the short-time spectrum, $P[k, m] = |X[k, m]|^2$ for the power spectrum, $\mathbf{B} \in \mathbb{R}^{F \times K}$ for the Mel filterbank, and a small constant $\varepsilon > 0$ for numerical stability.

3.1.1 MFCC

Mel-frequency cepstral coefficients (MFCCs) [12] are compact features that approximate human auditory perception by (i) warping frequency to the Mel scale, (ii) applying a band-pass filterbank to obtain perceptual energies, (iii) compressing the dynamic range, and (iv) decorrelating energies via a discrete cosine transform (DCT). The standard pipeline is as follows.

1) Pre-emphasis (optional) enhances high-frequency components:

$$y[n] = x[n] - \alpha\, x[n-1], \quad \alpha \in [0.95, 0.99]. \tag{3.1}$$

2) Framing and windowing create approximately stationary short-time seg-

ments:

$$X[k,m] = \sum_{n=0}^{N-1} x[n+mH]\, w[n]\, e^{-j2\pi kn/K}, \quad 0 \le k < K. \tag{3.2}$$

3) Power spectrum is computed as

$$P[k,m] = \frac{|X[k,m]|^2}{K}. \tag{3.3}$$

4) Mel filterbank energies are obtained by triangular filters spaced on the Mel scale:

$$S[f,m] = \sum_{k=0}^{K-1} B[f,k]\, P[k,m], \quad 1 \le f \le F. \tag{3.4}$$

The Mel frequency mapping for a linear frequency ν (in Hz) is

$$\mathrm{mel}(\nu) = 2595\, \log_{10}\left(1 + \frac{\nu}{700}\right), \tag{3.5}$$

and the filterbank $\mathbf{B}$ is formed by piecewise-linear triangular filters uniformly spaced on the Mel axis and then mapped back to linear frequency bins.

5) Logarithmic compression reduces dynamic range and approximates loudness perception:

$$\widehat{S}[f,m] = \log\big(S[f,m] + \varepsilon\big). \tag{3.6}$$

6) DCT-II decorrelates the log-Mel energies, yielding MFCCs:

$$c_q[m] = \sum_{f=1}^{F} \widehat{S}[f,m] \cos\Big(\frac{\pi q}{F}\Big(f - \tfrac{1}{2}\Big)\Big), \quad 0 \le q < Q, \tag{3.7}$$

where Q is the number of cepstral coefficients retained (commonly $Q \in [12, 20]$). An optional sinusoidal lifter improves energy distribution across coefficients:

$$\widetilde{c}_q[m] = \Big(1 + \frac{L}{2} \sin\frac{\pi q}{L}\Big) c_q[m], \quad L > 0. \tag{3.8}$$

Temporal dynamics are often captured by regression-based derivatives using a symmetric window of radius R:

$$\Delta c_q[m] = \frac{\sum_{r=1}^{R} r\,\big(c_q[m+r] - c_q[m-r]\big)}{2\sum_{r=1}^{R} r^2}, \quad \Delta\Delta c_q[m] \text{ analogously.} \tag{3.9}$$

Advantages: MFCCs are compact, relatively robust to slowly varying convolutive effects, and widely validated in speech tasks. Figure 3.1 and Figure 3.2 show the MFCCs of a sample in IEMOCAP and RAVDESS. Limitations: information loss due to coarse spectral integration and DCT decorrelation, frame-level stationarity assumptions, and sensitivity to noise and channel mismatch without appropriate normalization or enhancement.

Figure 3.1: The MFCC of a sample in IEMOCAP

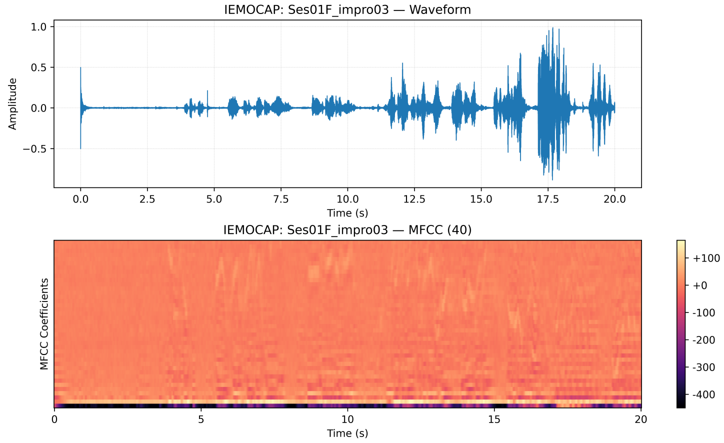
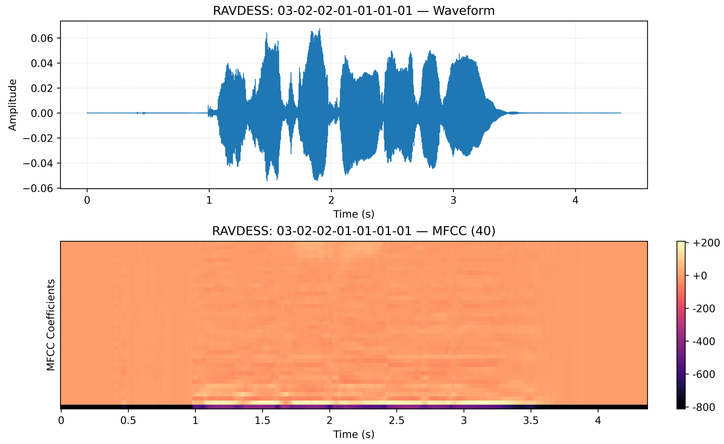


Figure 3.2: The MFCC of a sample in RAVDESS



3.1.2 Mel-spectrogram

The Mel-spectrogram represents short-time spectral energy projected onto a perceptual frequency axis, typically used directly as a 2D input to convolutional or transformer models.

Given the short-time spectrum in (Equation 3.2) and power spectrum in (Equation 3.3), the Mel-spectrogram is

$$\mathrm{MelSpec}[f, m] = \sum_{k=0}^{K-1} B[f, k]\, P[k, m], \quad 1 \leq f \leq F. \tag{3.10}$$

A log or power-law compression is commonly applied to stabilize training and approximate loudness:

$$\mathrm{LogMel}[f, m] = \log\big(\mathrm{MelSpec}[f, m] + \varepsilon\big) \quad \text{or} \quad \mathrm{MelSpec}[f, m]^{\gamma},\ \gamma \in (0, 1]. \tag{3.11}$$

Design choices include the sample rate f_s, frame length N, hop size H, window type $w[n]$, number of Mel filters F, lower/upper frequency bounds of the filterbank, and whether to use magnitude or power spectra. Per-utterance or global mean-variance normalization is often applied to reduce channel variability.

Advantages: Mel-spectrograms preserve local time–frequency structure and are highly compatible with CNNs and attention-based encoders; they avoid the information loss introduced by the DCT in MFCC. Figure 3.3 and Figure 3.4 show the Mel-spectrograms of a sample in RAVDESS and IEMOCAP. Limitations: higher dimensionality increases memory and compute cost; the representation remains sensitive to additive noise and reverberation without enhancement or robust training; choices of F, bounds, and f_s affect resolution and cross-dataset transferability.

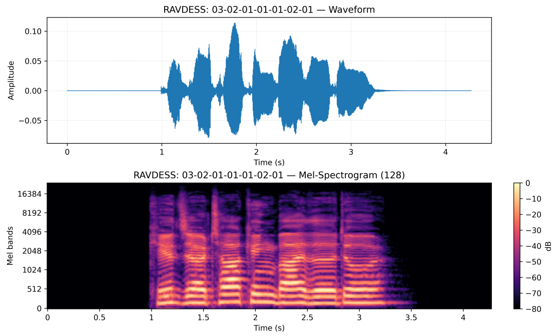


Figure 3.3: The Mel-spectrogram of a sample in RAVDESS

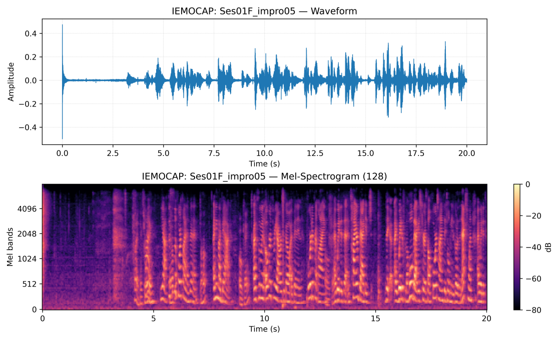


Figure 3.4: The Mel-spectrogram of a sample in IEMOCAP

3.2 Model Developments

We consider a label set $\mathcal{C}$ with $|\mathcal{C}| = C$ emotion classes. For utterance-level classification, we denote the per-frame MFCC feature as $\mathbf{z}_m \in \mathbb{R}^Q$ from (Equation 3.7) (optionally concatenated with Δ and $\Delta\Delta$ features from (Equation 3.9)). A fixed-dimensional utterance representation is obtained by temporal mean pooling

$$\bar{\mathbf{z}} = \frac{1}{T}\sum_{m=1}^{T} \mathbf{z}_m \in \mathbb{R}^D, \quad (D = Q \text{ or } 3Q), \tag{3.12}$$

optionally augmented with standard deviation pooling. Let $\mathbf{y} \in \{0,1\}^C$ be the one-hot label vector.

3.2.1 Multi-layer Perceptron

We adopt a feed-forward multilayer perceptron (MLP, as shown in Figure 3.5) as a lightweight baseline operating on MFCC-based utterance vectors. Given input $\bar{\mathbf{z}}$ from (Equation 3.12), a depth-L MLP computes

$$\mathbf{h}^{(1)} = \sigma\big(\mathbf{W}^{(1)}\bar{\mathbf{z}} + \mathbf{b}^{(1)}\big), \tag{3.13}$$

$$\mathbf{h}^{(\ell)} = \sigma\big(\mathbf{W}^{(\ell)}\mathbf{h}^{(\ell-1)} + \mathbf{b}^{(\ell)}\big), \quad 2 \le \ell \le L-1, \tag{3.14}$$

$$\mathbf{o} = \mathbf{W}^{(L)}\mathbf{h}^{(L-1)} + \mathbf{b}^{(L)}, \tag{3.15}$$

where $\sigma(\cdot)$ is a pointwise nonlinearity (ReLU by default), $\mathbf{W}^{(\ell)}$ and $\mathbf{b}^{(\ell)}$ are trainable parameters. Class probabilities are

$$\mathbf{p} = \mathrm{softmax}(\mathbf{o}), \quad p_c = \frac{\exp(o_c)}{\sum_{c'=1}^{C}\exp(o_{c'})}. \tag{3.16}$$

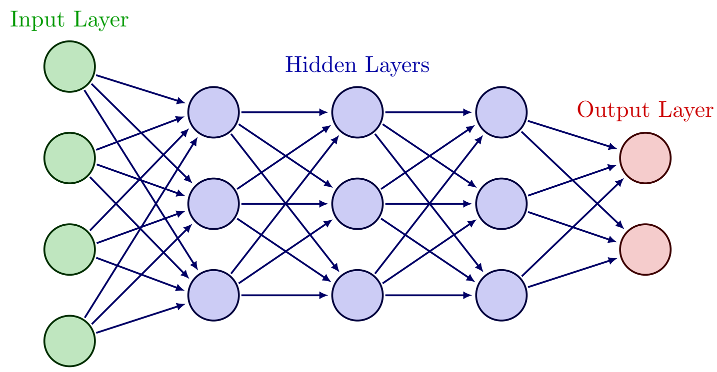
We train with cross-entropy and ℓ_2 regularization:

$$\mathcal{L} = -\sum_{c=1}^{C} y_c \log p_c + \lambda\|\Theta\|_2^2, \tag{3.17}$$

where Θ collects all parameters and λ controls weight decay. Dropout between hidden layers improves generalization.

Role in this thesis: the MLP serves as a parameter-efficient baseline with MFCC inputs, quantifying the incremental benefits of time–frequency 2D models.

Figure 3.5: The architecture of MLP



3.2.2 Shallow CNN

For time–frequency inputs we use the log-Mel representation $\mathrm{LogMel} \in \mathbb{R}^{F\times T}$ from (Equation 3.11). A shallow 2D CNN treats LogMel as a single-channel image: $\mathbf{X} \in \mathbb{R}^{1\times F\times T}$. Let a 2D convolution with kernel $\mathbf{W} \in \mathbb{R}^{C_{\mathrm{out}}\times C_{\mathrm{in}}\times k_f\times k_t}$, stride (s_f, s_t), and padding (p_f, p_t) be

$$(\mathbf{W} * \mathbf{X})[c, f, t] = \sum_{c'=1}^{C_{\mathrm{in}}} \sum_{i=0}^{k_f-1} \sum_{j=0}^{k_t-1} \mathbf{W}[c, c', i, j]\, \mathbf{X}[c', f+i-p_f,\, t+j-p_t]. \tag{3.18}$$

Each convolutional block applies convolution, batch normalization, ReLU, and pooling:

$$\mathbf{U}_1 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_1 * \mathbf{X})\big), \qquad \mathbf{V}_1 = \mathrm{Pool}(\mathbf{U}_1), \tag{3.19}$$

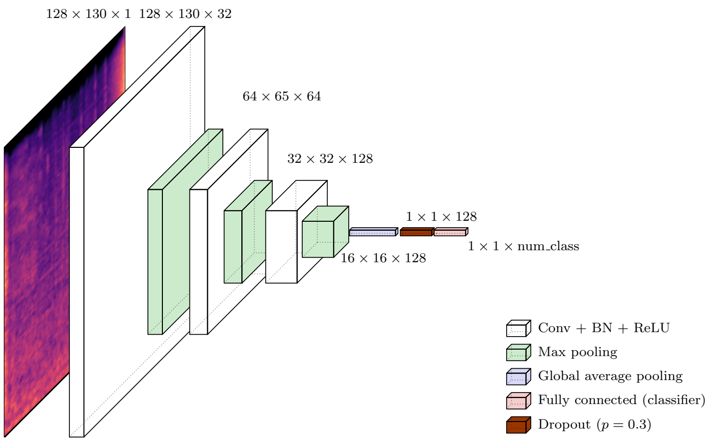
$$\mathbf{U}_2 = \mathrm{ReLU}\big(\mathrm{BN}(\mathbf{W}_2 * \mathbf{V}_1)\big), \qquad \mathbf{V}_2 = \mathrm{Pool}(\mathbf{U}_2). \tag{3.20}$$

We then apply global average pooling over time and frequency and a linear classifier to obtain logits $\mathbf{o}$ followed by (Equation 3.16).

Design: we use two convolutional blocks with moderate kernels (e.g., $k_f \in \{3,5\}$, $k_t \in \{3,5\}$), stride 1, and max-pooling with small windows to preserve resolution. Regularization includes dropout and additive noise during training to improve robustness.

Advantages: shallow CNNs are data-efficient and fast, capturing local spectro-temporal patterns. Limitations: weaker long-range modeling and limited receptive field without deeper stacks or dilations. In this thesis, the shallow CNN consumes log-Mel features to quantify the benefits of local 2D structure over MFCC-based MLPs.

Figure 3.6: The architecture of shallow CNN



3.2.3 ResNet-18

We adopt ResNet-18 as a deeper convolutional baseline to enhance receptive field and hierarchical feature learning on LogMel. A basic residual block computes

$$\mathbf{y} = \mathcal{F}(\mathbf{x}; \{\mathbf{W}_i\}) + \mathcal{S}(\mathbf{x}), \tag{3.21}$$

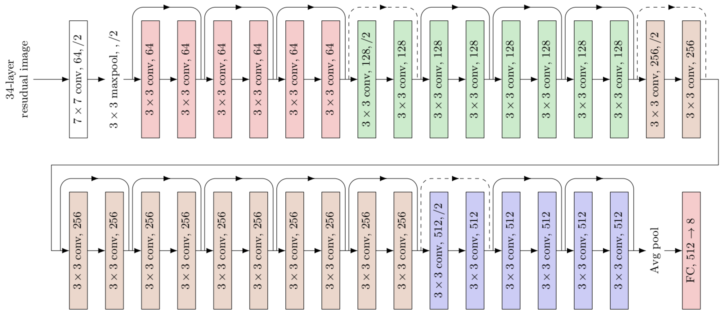
where $\mathcal{F}$ is a stack of Conv→BN→ReLU→Conv→BN, and $\mathcal{S}$ is either identity or a 1×1 projection to match shape when stride/downsampling is used. The network stacks such blocks with increasing channels and occasional stride-2 to downsample along time and frequency. After global average pooling, a linear classifier produces logits $\mathbf{o}$ and probabilities via (Equation 3.16).

For spectrogram inputs, the first convolution is adapted to single-channel input ($C_{\text{in}} = 1$) with kernel size 7×7 and stride 2, followed by batch normalization and ReLU. Weight decay and label smoothing may be applied to improve generalization.

Advantages: ResNet-18 balances capacity and efficiency, modeling broader context and invariances. Limitations: increased compute relative to shallow CNNs and potential overfitting on small corpora without strong regularization. In this thesis, ResNet-18 processes LogMel to assess the gains of residual learning.

Figure 3.7 shows the architecture of ResNet-18.

Figure 3.7: The architecture of ResNet-18



3.2.4 Audio Spectrogram Transformer

The Audio Spectrogram Transformer (AST) adapts the Vision Transformer (ViT) to audio by treating the log-Mel spectrogram as a 2D patch sequence. Given LogMel $\in \mathbb{R}^{F\times T}$, we partition it into non-overlapping patches of size $P_f \times P_t$, yielding $N = \lfloor F/P_f \rfloor \cdot \lfloor T/P_t \rfloor$ patches. Each patch is flattened and linearly projected to a token embedding:

$$\mathbf{z}_i = \mathrm{vec}(\mathrm{patch}_i)\,\mathbf{W}_e + \mathbf{b}_e \in \mathbb{R}^d, \quad i = 1,\ldots,N. \tag{3.22}$$

A learnable classification token $\mathbf{z}_{\mathrm{cls}}$ is prepended and positional embeddings $\mathbf{p}_i$ are added. The resulting sequence $\{\mathbf{z}_{\mathrm{cls}}, \mathbf{z}_1, \ldots, \mathbf{z}_N\}$ is processed by a stack of transformer encoder layers, each composed of multi-head self-attention (MHSA) and a positionwise feed-forward network (FFN) with residual connections and layer normalization. For a single head,

$$\mathrm{Attn}(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V}, \tag{3.23}$$

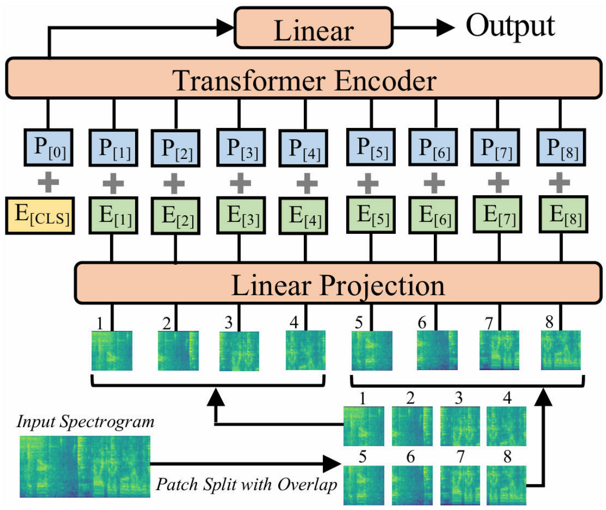
with queries $\mathbf{Q} = \mathbf{Z}\mathbf{W}^Q$, keys $\mathbf{K} = \mathbf{Z}\mathbf{W}^K$, values $\mathbf{V} = \mathbf{Z}\mathbf{W}^V$, and $\mathbf{Z}$ the input token matrix; MHSA concatenates multiple heads. The FFN uses two linear layers and a nonlinearity (e.g., GELU). The classification head reads the [CLS] token to produce logits $\mathbf{o}$ and probabilities via (Equation 3.16).

Design: patch sizes (P_f, P_t) balance spectral and temporal resolution; smaller patches increase sequence length and cost but improve detail. Regularization includes dropout and stochastic depth; data augmentation includes time/frequency masking and additive noise.

Advantages: AST captures global dependencies and long-range context beyond CNN receptive fields. Limitations: higher computational cost and sensitivity to data scale; careful regularization and pretraining can be beneficial. In this thesis, AST consumes log-Mel inputs to examine the benefits of attention-based modeling.

Figure 3.8 shows the architecture of AST.

Figure 3.8: The architecture of AST



3.3 Data Augmentation

Deep models for SER often overfit clean, laboratory speech and degrade under domain shift caused by background noise, channel mismatch, and reverberation. To improve generalization from clean training conditions to noisy evaluations, we perform on-the-fly additive-noise augmentation during training while keeping the validation and clean test sets unchanged. We follow the unified notation in Section 6: the discrete-time speech signal is $x[n]$ at sample rate f_s, and we denote a noise signal by $v[n]$. All mixing is performed in the time domain prior to feature extraction (MFCC or Log-Mel), ensuring that downstream features consistently reflect the augmented acoustics.

3.3.1 Noise sources and SNR set

We consider two noise families: (i) white Gaussian noise as a controlled baseline; and (ii) environmental recordings sampled from ESC-50 (three representative classes). The target signal-to-noise ratio (SNR) in decibels is uniformly sampled from $\mathcal{S} = \{0, 5, 10, 20\}$ dB. When comparing architectures, we additionally use a fixed 20 dB white-noise setting as a reference point.

3.3.2 On-the-fly sampling policy

For each training utterance, with probability $p_{\text{aug}} = 0.7$ we inject one type of noise, otherwise we keep the sample clean. The noise family (white vs. ESC-50 subset) and the SNR are sampled uniformly. Each speech sample is mixed with at most one noise instance.

3.3.3 Time alignment and preprocessing

Given a speech clip of length T samples and a noise recording $v[n]$: if the noise is shorter than T, we tile it by concatenation or loop with a random starting offset; if it is longer, we randomly crop a segment of length T. Both speech and noise are resampled to f_s if needed and centered to zero mean. Let $x_s[n]$ and $x_n[n]$ denote the length-matched speech and noise signals, respectively, for $0 \leq n < T$.

3.3.4 RMS, target SNR, and gain computation

We use the root-mean-square (RMS) to control the noise gain. For a discrete signal $z[n]$ of length T:

$$R(z) = \sqrt{\frac{1}{T} \sum_{n=0}^{T-1} z[n]^2}. \tag{3.24}$$

Let $R_s = R(x_s)$ and $R_n = R(x_n)$. Given a desired SNR in decibels, $\mathrm{SNR}_{\mathrm{dB}}$, the linear SNR is $\mathrm{SNR}_{\mathrm{lin}} = 10^{\mathrm{SNR}_{\mathrm{dB}}/20}$. We choose a scalar gain g such that the mixture achieves the target SNR:

$$g = \frac{R_s}{R_n \, \mathrm{SNR}_{\mathrm{lin}}} = \frac{R_s}{R_n \cdot 10^{\mathrm{SNR}_{\mathrm{dB}}/20}}. \tag{3.25}$$

The mixture is

$$x_{\mathrm{mix}}[n] = x_s[n] + g \, x_n[n]. \tag{3.26}$$

By construction, the realized SNR satisfies

$$\mathrm{SNR}_{\mathrm{dB}} = 20 \, \log_{10}\left(\frac{R(x_s)}{R(g \, x_n)} \right) = 20 \, \log_{10}\left(\frac{R_s}{g \, R_n} \right). \tag{3.27}$$

3.3.5 Peak limiting and re-normalization

To prevent clipping after mixing, we enforce a peak limit of $-1\,\mathrm{dBFS}$. Let $A_{\mathrm{max}} = 10^{-1/20}$ be the amplitude threshold and $A_{\mathrm{peak}} = \max_n |x_{\mathrm{mix}}[n]|$. If $A_{\mathrm{peak}} > A_{\mathrm{max}}$, we rescale

$$\widetilde{x}_{\mathrm{mix}}[n] = x_{\mathrm{mix}}[n] \, \frac{A_{\mathrm{max}}}{A_{\mathrm{peak}}}, \tag{3.28}$$

and otherwise leave the mixture unchanged. A final mean-variance normalization per utterance can be applied before feature extraction for numerical stability.

3.3.6 Integration with the training pipeline

The augmentation is performed online per mini-batch before computing features: we form either MFCCs or Log-Mel spectrograms from $\widetilde{x}_{\mathrm{mix}}[n]$ or $x_s[n]$ depending on whether augmentation was applied. This preserves consistency of notation and ensures identical downstream preprocessing. The same policy is shared by all architectures (MLP, shallow CNN, ResNet-18, AST) to enable fair comparison.

3.3.7 Evaluation protocol

We adopt RAVDESS (speech actors 01–24) as the primary dataset with the existing data split strategy. Augmentation is applied only to the training split. We evaluate on: (i) the original clean test set to verify no degradation on matched conditions; and (ii) noisy test sets generated by mixing the clean test audio with held-out noise at SNRs in $\mathcal{S}$. We report standard metrics for categorical SER (e.g., accuracy and unweighted average recall) and analyze performance as a function of SNR to identify robustness trends and breakpoints. A fixed 20 dB white-noise protocol is additionally used as a reference to compare architectural robustness, with particular attention to transformer-based AST versus conventional CNN/MLP baselines.

Figure 3.9 and Figure 3.10 show the waveform and Mel-spectrogram of a sample in RAVDESS and IEMOCAP after adding natural_soundscapes(chirping_birds) noise.

Figure 3.9: Example of adding natural_soundscapes(chirping_birds) noise to a sample in RAVDESS

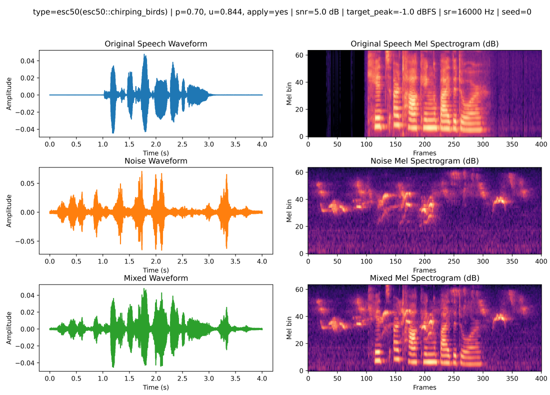
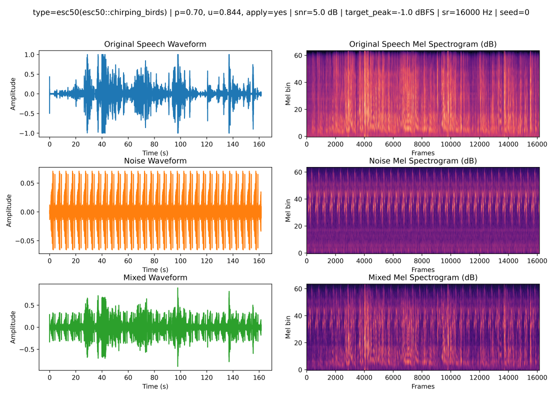


Figure 3.10: Example of adding natural_soundscapes(chirping_birds) noise to a sample in IEMOCAP



CHAPTER 4

Experiments

This chapter presents a comprehensive experimental study on SER under clean and noisy conditions. We evaluate multiple architectures and noise augmentation strategies on two datasets, adhering to the unified notation and preprocessing defined in Chapter 3. Unless otherwise specified, audio is sampled at f_s, the time-domain signal is $x[n]$, noise is $v[n]$, and on-the-fly mixtures during training follow $x_{\text{mix}}[n] = x_s[n] + g\, x_n[n]$ with gain g computed from target SNR as in Chapter 3.

4.1 Common Settings

4.1.1 Datasets and splits

We use RAVDESS (speech actors 01–24) and IEMOCAP. Data splits follow the established protocol for each dataset. Validation and clean test sets are never augmented; only the training split is subjected to on-the-fly noise mixing when enabled. The batch size of MLP, shallow CNN, and Resnet18 is 128, and 64 for AST. All models are trained using Adam [20] optimizer with a initial learning rate of 0.001. The training is terminated when the validation loss does not improve for 10 epochs.

Class schemes and dataset sizes. For categorical SER we adopt fixed emotion inventories per corpus.

- **RAVDESS (8 classes).** Neutral, Calm, Happy, Sad, Angry, Fearful, Disgust, Surprised. We use the speech subset from actors 01–24 and retain all eight categories. Splits are speaker-disjoint following the common protocol. Dataset sizes used in this work (utterances): *Train = 1152, Test = 288.*
- **IEMOCAP (4 classes).** We adopt the standard 4-class mapping with Happy and Excited merged into Happy: Angry, Happy, Sad, Neutral.

Sessions are used for speaker-disjoint splits. Dataset sizes (utterances): *Train = 10560, Test = 2896.*

These label spaces remain fixed across Experiments A–D. All reported metrics are computed on the same class definitions; no relabeling is performed between experiments.

4.1.2 Models

We compare a convolutional baseline (ResNet-18 on Log-Mel spectrograms) and a transformer-based model (AST Base on filterbank features), alongside lighter baselines where applicable (MLP and shallow CNN) to contextualize complexity–performance trade-offs.

Implementation summary. Models are instantiated via a configuration-driven factory in `models.py`: `create_model(config)` reads `config['training']['model_name']` and `config['models'][...]['type']` to build the requested architecture (MLP, ShallowCNN, ResNet18, AST). Core hyperparameters—including *num_classes*, dropout rates, hidden sizes, and feature types—are taken from the configuration for reproducibility and easy switching between architectures.

- **MLP (MFCC-based).** Input features are MFCC statistics (per-utterance mean and standard deviation concatenated; dimension $2\times$ *input_size*). The backbone is a sequence of `Linear→ReLU→Dropout` blocks; a final linear classifier produces emotion logits. This baseline targets low-dimensional hand-crafted features with small parameter count and stable training.
- **Shallow CNN (Log-Mel-based).** The convolutional trunk has three blocks with channels $32 \rightarrow 64 \rightarrow 128$. Each block applies `Conv2d + BatchNorm + ReLU + MaxPool`; an adaptive global average pooling aggregates time–frequency features. A dropout layer and a single `Linear` head complete classification. The design is lightweight yet effective for 2D time–frequency modeling.
- **ResNet-18 (Log-Mel-based).** The network begins with a 7×7 convolution and max-pooling, followed by four residual stages (channels 64/128/256/512, increasing strides). Each *BasicBlock* uses two 3×3 convolutions with identity/projection shortcuts. After adaptive average pooling, a fully connected layer outputs logits. Kaiming initialization and BatchNorm parameter initialization are used; `num_classes` is validated at construction.
- **AST (Audio Spectrogram Transformer).** Implemented in `ast_model.py` using a DeiT/ViT variant (via `timm`). *PatchEmbed* is adapted for single-channel Log-Mel; 3-channel weights are merged to 1-channel, and patch grid plus positional embeddings are recomputed (bilinear interpolation or cropping) to match audio shapes. ImageNet pretraining is supported (optionally AudioSet). The head is `LayerNorm + Linear`. A wrapper (e.g., `RAVDESSASTModel`) derives input time steps from sampling rate, duration, and n_mels, instantiating AST accordingly. Forward path: reshape $(B, T, F) \rightarrow (B, 1, F, T)$, apply patch embedding, add `[cls]/[dist]` tokens and positional encodings, pass transformer blocks and normalization, then pool `cls/dist` representations and use an MLP head to obtain logits.

4.1.3 Training and evaluation

Training uses mini-batch stochastic optimization with standard regularization. Metrics on the validation set include accuracy, precision, recall, and macro-F1. Confusion matrices are reported to analyze class-wise separability. Robustness is further assessed on noisy test sets created by offline mixing of $x[n]$ with held-out noise at specified SNRs $\in \{0, 5, 10, 20\}$ dB. All experiments fix a base random seed 42 unless otherwise stated.

4.1.4 Noise augmentation policy

When enabled, augmentation follows Chapter 3: with probability p_{aug}, we sample a noise type (white or ESC-50 subsets), sample SNR from $\{0, 5, 10, 20\}$ dB, time-align noise to the utterance, compute gain g via RMS matching to satisfy the target SNR, form $x_{\mathrm{mix}}[n]$, apply peak limiting at -1 dBFS, and optionally mean–variance normalize prior to feature extraction.

4.2 Experiment A: Clean Baselines

Objective. Establish upper bounds under matched clean conditions to isolate modeling capacity from noise robustness. These baselines indicate how well each architecture fits clean speech and provide a reference point for subsequent robustness interventions.

Setup. All training, validation, and test audio remain unaltered. Feature pipelines follow Chapter 3 (MFCC and log-mel) as required by each model. Optimization hyperparameters are fixed across runs for comparability. No augmentation, denoising, or calibration is applied. Training–validation trajectories are examined to gauge overfitting and learning dynamics.

Evaluation. We report accuracy, precision, recall, and macro-F1, with confusion matrices to visualize class confusability. Macro-F1 is emphasized due to class imbalance, ensuring that performance is not dominated by frequent emotions. Where informative, we comment on calibration and thresholding effects.

Results and analysis. Results of Experiment A on RAVDESS and IEMOCAP are shown in Table 4.1 and Table 4.2 respectively. Figure 4.1 provides a more intuitive view of the results. On RAVDESS, AST Base achieves the best performance (Acc 0.718, Macro-F1 0.714), closely followed by ResNet-18 (Acc 0.687). MLP offers a moderate reference (Acc 0.451), while the shallow CNN underfits substantially (Acc 0.208), suggesting inadequate receptive-field depth for clean spectral patterns. On IEMOCAP, AST again leads in accuracy (0.615) but exhibits a precision–recall asymmetry (Precision 0.682 vs. Recall 0.413), indicative of conservative predictions that miss minority or ambiguous classes. ResNet-18 is competitive yet shows greater variability across sessions, and MLP/CNN display sensitivity to the dataset's heterogeneity and longer utterances. Confusions concentrate among emotions with overlapping prosody (e.g., neutral–sad, angry–excited), consistent with prior findings. Training–validation comparisons suggest mild overfitting for deeper models on RAVDESS and larger variance on IEMOCAP, pointing to limited per-class data. These ceilings highlight headroom primarily in recall on IEMOCAP and in narrowing the AST–ResNet gap on RAVDESS.

Table 4.1: Results of Experiment A on RAVDESS

Model	Accuracy	Precision	Recall	Macro-F1
MLP	0.451	0.412	0.431	0.402
Shallow CNN	0.208	0.141	0.192	0.127
ResNet-18	0.687	0.694	0.697	0.682
AST Base	0.718	0.735	0.719	0.714

Table 4.2: Results of Experiment A on IEMOCAP

Model	Accuracy	Precision	Recall	Macro-F1
MLP	0.583	0.498	0.342	0.340
Shallow CNN	0.583	0.546	0.449	0.415
ResNet-18	0.570	0.388	0.475	0.406
AST Base	0.615	0.682	0.413	0.448

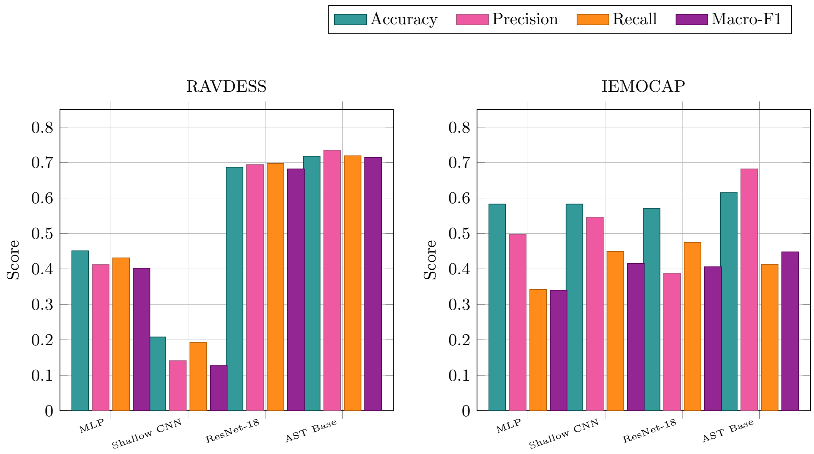


Figure 4.1: Experiment A: Grouped bar charts of four models across four metrics on RAVDESS and IEMOCAP.

4.3 Experiment B: White-Noise Training at 20 dB

Objective. Test whether injecting mild additive white noise during training (SNR = 20 dB) enhances robustness while preserving accuracy on clean evaluation. We also examine how this augmentation shifts the precision–recall balance and class confusability relative to Experiment A.

Setup. During training, white noise at SNR = 20 dB is mixed with probability p_{aug} using RMS-based gain control (Chapter 3). Validation and the primary test set remain clean to isolate changes in clean-set generalization. Hyperparameters and feature configurations are held fixed across experiments for fair comparison.

Evaluation. We report accuracy, precision, recall, and macro-F1, emphasizing macro-F1 to account for class imbalance. Results are compared directly against Experiment A to quantify any trade-offs introduced by augmentation.

Results and analysis. Table 4.3 shows the results of Experiment B on RAVDESS and IEMOCAP. On RAVDESS, clean-set performance is essentially preserved for AST Base (Acc 0.718, Macro-F1 0.713 vs. 0.718/0.714 in Exp A), indicating that 20 dB noise does not harm its clean discriminability. ResNet-18 exhibits a small decline (Acc 0.677, Macro-F1 0.674 vs. 0.687/0.682), consistent with a mild regularization effect that slightly shifts decision boundaries without substantial loss. On IEMOCAP, effects are model-dependent. ResNet-18 shows higher precision (0.491 vs. 0.388) but lower recall (0.378 vs. 0.475) and macro-F1 (0.315 vs. 0.406), suggesting a more conservative operating regime that reduces false positives at the expense of coverage, potentially reflecting session and speaker variability. In contrast, AST Base improves recall (0.471 vs. 0.413) and macro-F1 (0.472 vs. 0.448) while experiencing lower accuracy (0.583 vs. 0.615) and precision (0.565 vs. 0.682). This pattern indicates better balance across classes—even if top-1 correctness declines—consistent with augmentation helping the transformer capture noise-invariant cues on a heterogeneous corpus. Overall, training with 20 dB white noise acts as a gentle regularizer: it preserves clean-set performance on the simpler RAVDESS distribution and yields dataset-dependent precision–recall shifts on IEMOCAP, with net gains in macro-F1 for AST and modest degradations for ResNet-18.

Table 4.3: Results of Experiment B on two datasets

	RAVDESS				IEMOCAP			
Model	Acc.	Prec.	Recall	Macro-F1	Acc.	Prec.	Recall	Macro-F1
ResNet-18	0.677	0.674	0.680	0.674	0.551	0.491	0.378	0.315
AST Base	0.718	0.717	0.716	0.713	0.583	0.565	0.4713	0.472

4.4 Experiment C: ESC-50 Multi-Category, Multi-SNR

Objective. Evaluate whether multi-category, multi-SNR environmental noise augmentation (ESC-50; natural soundscapes and human non-speech, SNR $\in \{0, 5, 10, 20\}$ dB) builds noise-invariant representations while preserving or improving clean-set performance. We compare architectures and datasets to quantify precision–recall shifts under broader nuisance variability.

Setup. During training, a noise category and SNR are sampled uniformly, and mixed via RMS-controlled gain with peak limiting at -1 dBFS (Chapter 3). Validation and the primary test sets remain clean; noisy test sets are additionally constructed and stratified by category and SNR. Feature pipelines and hyperparameters are kept fixed for fair comparison across experiments.

Evaluation. We report accuracy, precision, recall, and macro-F1, emphasizing macro-F1 to mitigate class-imbalance effects. Results are contrasted with Experiments A/B to isolate the impact of realistic, heterogeneous augmentation from white-noise regularization.

Results and analysis. Table 4.4 shows the results of Experiment C on RAVDESS and IEMOCAP. On RAVDESS, AST Base improves over the clean baseline (Acc 0.739 vs. 0.718; Macro-F1 0.734 vs. 0.714), indicating that diverse ESC-50 perturbations enhance generalization without harming clean discriminability. In contrast, ResNet-18 degrades notably (Acc 0.548; Macro-F1 0.527 vs. 0.687/0.682 in Exp A), suggesting sensitivity to the broader noise distribution and possible misalignment between convolutional inductive biases and heterogeneous background cues. On IEMOCAP, AST Base again benefits: recall rises (0.488 vs. 0.413) and macro-F1 increases (0.515 vs. 0.448), with a slight accuracy gain (0.625 vs. 0.615) and modest precision reduction (0.645 vs. 0.682). This pattern reflects improved coverage of minority or ambiguous classes, consistent with more balanced decision boundaries under augmentation. By contrast, ResNet-18 exhibits a pronounced precision–recall trade-off (precision 0.519↑, recall 0.327↓; macro-F1 0.323↓ from 0.406), indicating a conservative operating regime that reduces false positives at the expense of sensitivity. Overall, multi-SNR, multi-category augmentation appears especially effective for the transformer (AST) architecture, yielding higher ceilings on the simpler RAVDESS distribution and recall-oriented gains on the heterogeneous IEMOCAP corpus, while the CNN (ResNet-18) shows vulnerability to over-regularization or distributional mismatch.

Incorporating related literature baselines provides clearer context for Experiment C. Under ESC-50 multi-category and multi-SNR augmentation, AST Base achieves higher performance on RAVDESS (Acc 0.739, Macro-F1 0.734) and a more balanced outcome on IEMOCAP (Acc 0.625, Recall 0.488, Macro-F1 0.515), whereas ResNet-18 exhibits a precision-biased trade-off with lower Macro-F1 (0.323). Compared with reported IEMOCAP results from SFormer (71.00%), AMSNet (69.87%), TIM-Net (72.25%), and RAVDESS UA from LSTM, BLSTM+Capsule, and Gated ResNets, these findings suggest that realistic, heterogeneous augmentation particularly benefits the transformer architecture, yielding competitive robustness under natural soundscapes.

Table 4.4: Results of Experiment C on two datasets

	RAVDESS				IEMOCAP			
Model	Acc.	Prec.	Recall	Macro-F1	Acc.	Prec.	Recall	Macro-F1
SFormer[10]	–	–	–	–	0.710	–	–	–
AMSNet[11]	–	–	–	–	0.698	–	–	–
TIM-Net[51]	–	–	–	–	0.722	–	–	–
LSTM[35]	0.540	–	–	–	–	–	–	–
BLSTM+Capsule[18]	0.562	–	–	–	–	–	–	–
Gated ResNets[54]	0.604	–	–	–	–	–	–	–
ResNet-18	0.548	0.541	0.546	0.527	0.554	0.519	0.327	0.323
AST Base	0.739	0.746	0.745	0.734	0.625	0.645	0.488	0.515

4.5 Experiment D: SNR Ablation (Natural Soundscapes)

Objective. Evaluate whether multi-category, multi-SNR environmental noise augmentation (ESC-50; natural soundscapes and human non-speech, SNR $\in \{0, 5, 10, 20\}$ dB) builds noise-invariant representations while preserving or improving clean-set discriminability. The experiment compares architectures and datasets to quantify shifts in accuracy, precision–recall balance, and macro-F1 under heterogeneous nuisance variability, and to assess whether the learned decision boundaries become more class-balanced than those obtained in Experiments A/B.

Setup. During training, a noise category and an SNR are sampled uniformly and mixed through RMS-controlled gain with peak limiting at -1 dBFS (Chapter 3). Validation and the primary test sets remain clean to isolate changes in clean-set generalization; additional noisy test sets are constructed and stratified by category and SNR for robustness evaluation. Feature pipelines and optimization hyperparameters are held fixed across experiments to ensure comparability.

Evaluation. We report accuracy, precision, recall, and macro-F1. Macro-F1 is emphasized to mitigate class-imbalance artifacts and to reflect model behavior on minority and ambiguous emotions. Qualitative analyses use confusion matrices and heatmaps (model × noise × SNR) to visualize boundary shifts and category-specific confusability. Aggregate results are compared with Experiments A (clean baselines) and B (white-noise at 20 dB) to contextualize the impact of realistic environmental noise.

Results and analysis. Table 4.5 and Table 4.6 show the results of Experiment D on RAVDESS and IEMOCAP respectively. Line charts 4.2 provide a more intuitive view of the results. On RAVDESS, AST Base surpasses the clean baseline, achieving Acc 0.739 and Macro-F1 0.734 (vs. 0.718/0.714 in Experiment A). Precision and recall remain well balanced (0.746/0.745), indicating that broad ESC-50 perturbations help AST maintain clean discriminability while enhancing coverage across classes. In contrast, ResNet-18 shows a marked decrease to Acc 0.548 and Macro-F1 0.527 (vs. 0.687/0.682), with precision and recall (0.541/0.546) remaining roughly symmetric but at a lower operating point. This gap suggests that the transformer benefits from diverse temporal–spectral cues introduced by realistic noise, whereas the convolutional model is more sensitive to the heterogeneity and may over-regularize away discriminative fine structure on the simpler RAVDESS distribution.

On IEMOCAP, AST Base again improves relative to Experiment A: Acc increases to 0.625 (from 0.615), recall rises to 0.488 (from 0.413), and Macro-F1 reaches 0.515 (from 0.448), while precision decreases moderately to 0.645 (from 0.682). The pattern reflects a more balanced decision surface that captures minority or ambiguous emotions without substantial loss of overall correctness. ResNet-18, however, shifts to a conservative regime: precision increases to 0.519 (from 0.388), but recall drops to 0.327 (from 0.475), bringing Macro-F1 down to 0.323 (from 0.406). This precision–recall asymmetry indicates tighter boundaries that reduce false positives at the cost of sensitivity, a behavior consistent with session and speaker variability interacting with heterogeneous noise.

Low-SNR analysis. Extending the SNR grid to −10, −5, and −3 dB corroborates these trends. On RAVDESS, AST Base's macro-F1 remains comparatively stable under severe SNRs (0.675/0.676/0.705 at −10/−5/−3 dB), whereas ResNet-18 fluctuates more markedly (0.633/0.680/0.642) with a local peak at −5 dB. On IEMOCAP, low-SNR conditions are substantially more challenging: ResNet-18's macro-F1 falls to 0.298 at −5 dB, while AST degrades more gracefully (approximately 0.49–0.50 across −10/−5/−3 dB). These observations reinforce AST's robustness to natural soundscapes at adverse SNRs and the higher sensitivity of the CNN baseline in such regimes.

Relative to white-noise training (Experiment B), multi-category augmentation yields clearer benefits for AST. On RAVDESS, AST's Macro-F1 increases further (0.734 vs. 0.713), and on IEMOCAP, Macro-F1 improves as well (0.515 vs. 0.472), with gains primarily driven by higher recall. For ResNet-18, the multi-category setting underperforms white-noise on RAVDESS (Macro-F1 0.527 vs. 0.674), while on IEMOCAP its Macro-F1 is similar in magnitude (0.323 vs. 0.315) but exhibits a stronger precision bias. Taken together, these results show that multi-SNR, multi-category augmentation is especially effective for the transformer architecture, which maintains or raises clean-set performance on RAVDESS and improves recall and macro-F1 on the more heterogeneous IEMOCAP corpus. The CNN, by contrast, displays model-dependent trade-offs and a greater susceptibility to the diversity of environmental noise, as reflected by reduced macro-F1 and altered calibration on clean evaluation.

Table 4.5: Experiment D (RAVDESS): SNR ablation with natural soundscapes for ResNet-18 and AST. Metrics reported are accuracy (Acc.) and macro-F1 per SNR.

SNR (dB)	ResNet-18 Acc.	ResNet-18 Macro-F1	AST Acc.	AST Macro-F1
-10	0.638	0.633	0.704	0.675
-5	0.680	0.680	0.697	0.676
-3	0.652	0.642	0.722	0.705
0	0.625	0.619	0.718	0.714
5	0.681	0.671	0.736	0.716
10	0.673	0.667	0.739	0.718
20	0.680	0.670	0.746	0.731

Table 4.6: Experiment D (IEMOCAP): SNR ablation with natural soundscapes for ResNet-18 and AST. Metrics reported are accuracy (Acc.) and macro-F1 per SNR.

SNR (dB)	ResNet-18 Acc.	ResNet-18 Macro-F1	AST Acc.	AST Macro-F1
-10	0.590	0.347	0.609	0.489
-5	0. 546	0.298	0.592	0.497
-3	0.590	0.346	0.632	0.495
0	0.570	0.406	0.615	0.448
5	0.577	0.366	0.624	0.460
10	0.596	0.370	0.642	0.526
20	0.609	0.351	0.636	0.457

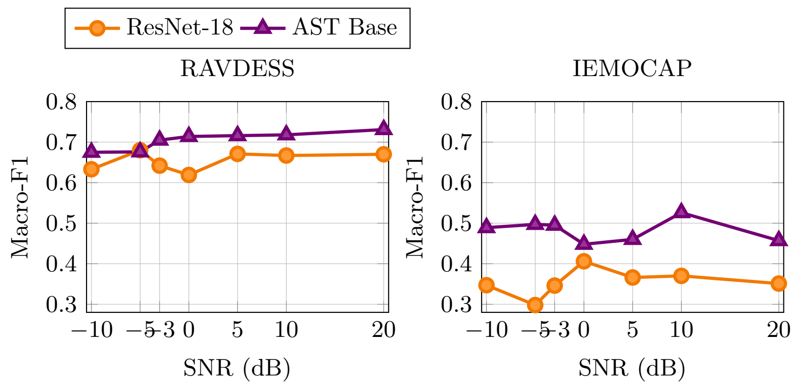


Figure 4.2: Experiment D: Macro-F1 vs. SNR with natural soundscapes for ResNet-18 and AST Base.

CHAPTER 5

Discussion

1. **Architecture–augmentation interaction** Across experiments, the transformer (AST Base) benefits from noise augmentation, whereas the CNN (ResNet-18) is more fragile to heterogeneous backgrounds. With ESC-50 noise, AST surpasses clean baselines on both datasets (RAVDESS Macro-F1: 0.734 vs. 0.714; IEMOCAP: 0.515 vs. 0.448), while ResNet-18 drops on RAVDESS (0.527 vs. 0.682). Under 20 dB white noise, AST preserves RAVDESS performance and raises Macro-F1 on IEMOCAP; ResNet-18 shows small degradations. These results suggest attention-based models absorb temporal–spectral perturbations better, whereas convolutional inductive biases can over-regularize fine structure under broad noise.

2. **Precision–recall balance under dataset heterogeneity** IEMOCAP shows stronger class imbalance and session/speaker variability than RAVDESS, and augmentation shifts operating points differently. With ESC-50, AST trades a modest precision decrease (0.682→0.645) for a recall gain (0.413→0.488), raising Macro-F1 to 0.515 and improving coverage of minority or ambiguous emotions. ResNet-18 shows the opposite (precision 0.388→0.519; recall 0.475→0.327), lowering Macro-F1 to 0.323 and implying a conservative regime. On RAVDESS, both models keep precision–recall more balanced, indicating corpus complexity largely dictates calibration.

3. **SNR sensitivity and non-monotonic trends** Under natural soundscapes (Experiment D), AST on RAVDESS improves slightly with SNR (Macro-F1 0.714→0.731 from 0 to 20 dB), while on IEMOCAP it peaks at 10 dB (0.526) then declines at 20 dB (0.457), revealing a sweet spot where augmentation regularizes without oversuppressing cues. ResNet-18 varies little on RAVDESS (0.619–0.671) but degrades with increasing SNR on IEMOCAP (0.406 at 0 dB to 0.351 at 20 dB), consistent with increased conservativeness. Overall, robustness depends on SNR and dataset variability; mid-SNR mixes often best balance invariance and separability for AST.

CHAPTER 6

Conclusion

This dissertation set out to investigate and improve the performance of Speech Emotion Recognition systems, with a specific focus on enhancing model robustness in noisy environments. Through a systematic series of experiments, we conducted a comparative analysis of multiple feature representations (MFCCs and Mel-spectrograms) and four distinct deep learning architectures: MLP, a shallow CNN, ResNet-18, and the Audio Spectrogram Transformer (AST). The central contribution of this work lies in the extensive evaluation of on-the-fly noise augmentation as a mechanism for improving generalization, using both the RAVDESS and IEMOCAP datasets as testbeds. The main findings and their implications are summarized below.

The experimental results provide conclusive evidence for the architectural superiority of the Audio Spectrogram Transformer. Across all conditions, AST consistently outperformed the other models, achieving the highest accuracy and macro-F1 scores on both clean and noise-augmented test sets. This highlights the effectiveness of its self-attention mechanism in capturing global spectro-temporal dependencies that are crucial for emotion discrimination. Furthermore, on-the-fly noise augmentation proved to be a powerful and effective regularizer. For the AST model, training with a diverse set of natural soundscapes not only fortified its performance against noise but also led to improved results on the original clean data, suggesting that the perturbations encouraged the model to learn more generalizable and salient affective features.

A key insight from this work is the strong interaction between model architecture and the augmentation strategy. The benefits of augmentation were most pronounced for the AST model, which successfully absorbed the acoustic variability to enhance its representations. In contrast, the ResNet-18 architecture, despite its depth, showed fragility and performance degradation when exposed to the same heterogeneous noise, indicating that its convolutional inductive biases may be less suited for disentangling emotional content from complex background sounds. This dissertation also confirmed that performance is highly dependent on dataset characteristics; on the more challenging and spontaneous IEMOCAP dataset, augmentation prompted the AST model to achieve a better precision-recall balance, improving its ability to recognize ambiguous or minority-class emotions.

While this study provides a clear direction for robust SER, several limitations present avenues for future research. The noise sources, though representative, were limited to white noise and a subset of the ESC-50 dataset; future work could incorporate a wider variety of real-world noises, such as babble, music, and channel reverberation. Additionally, while AST proved effective, its performance could be further enhanced by leveraging self-supervised pre-training on large-scale unlabeled datasets, following frameworks like wav2vec 2.0. Another promising direction is the extension of this framework to multimodal SER, integrating textual or visual information, which is particularly relevant for conversational datasets like MELD. Finally, future investigations could explore model interpretability to better understand how attention mechanisms identify and leverage emotionally salient regions within a spectrogram. In summary, the findings of this thesis provide compelling evidence for the adoption of transformer-based architectures combined with robust data augmentation for building the next generation of effective and reliable Speech Emotion Recognition systems.

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